



OPERATION Department
SIMHADRI



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LMI on "Operation & Maintenance Procedure for Dry Ash Evacuation & Transportation System"
LMI/OGN/OPS/SYST/044

Rev-01 Dec 2022 - LMI has been revised as per periodical review of 2 years period.

No modification made in this revision.

Put up for kind approval.

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Local Management Instructions

Operation & Maintenance Procedure for Dry Ash Evacuation & Transportation System



Approved for implementation by HEAD OF PROJECT

Date 12.12.2022

Prepared by OPERATION DEPARTMENT

NTPC Limited

Simhadri Thermal Power Project

LMI	<i>Operation & Maintenance Procedure for Dry Ash Evacuation & Transportation System</i>		
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OPERATION & MAINTENANCE PROCEDURE FOR DRY ASH EVACUATION & TRANSPORTATION SYSTEM

1.0	INTRODUCTION
	In Thermal Power Plants, Coal is burnt to generate power. Coal is used in Simhadri STPP of NTPC is having a very high ash content, generally in the range of 35 to 45%. The large amount of ash that is produced every day needs to be transported reliably, economically and in an environment-friendly manner. Around 80% of ash that is generated, the major portion, is carried by flue gases and referred as fly ash.
	Dry fly Ash Evacuation System (DAES) were introduced in 90s with the aim of reducing the water requirement in handling of fly ash. With the development of avenues for utilization of fly ash in cement, ash brick and FAB industries, second round of development in DAES system took place and NTPC started handling fly ash in completely dry form and started storing in fly ash silos from where user industries were offered dry fly ash for end use. Till then many of the plants were installed for the partial use of dry ash due to low demand of fly ash from the user industries. However, with changes in environmental norms in the country regarding the restriction in use of land to store waste material and water consumption, and reduction of environmental hazards due to coal ash, 100% utilization of power plant ash has been enforced.
	In Simhadri coal fired units, ash is removed in four different locations at Boiler, Bottom Ash hopper, Economizer hoppers, Air Pre Heater hoppers and ESP hoppers. Ash is transported from BAH, APH hoppers & ECO hoppers through slurry system and ESP hoppers through dry pneumatic System / wet system. Dry ash is transported in two phases, a) ESP hoppers to Buffer Hopper by vacuum extraction system and b) Buffer Hopper to Silo by pressure transportation.
	In St-1 Simhadri units are provided with 50% DAES facility and 100% stand by wet evacuation system. To achieve 100% ash utilization DAE augmentation system is being commissioned. St-2 is provided with 100% DAES facility with stand by 100% wet evacuation system.
2.0	SUPERSEDED DOCUMENTS & REVISION NOTES - Nil
	Revision 01: LMI has been revised as per 2 year periodical review. No changes made in this document.
3.0	SCOPE:
	This LMI document is applicable for 4 X 500 MW Units of Simhadri SMTTP, Stage-I & II units. The purpose of this LMI is to provide operation and maintenance personnel with guidelines for safe and effective operation and maintenance practices for system components associated with fly ash collection, handling, transport, and disposal.

Operation & Maintenance Procedure for Dry Ash Evacuation & Transportation System

	The scope covers equipment typically installed in the following general types of fly ash conveying systems: • Fly ash pressure conveying systems • Fly ash vacuum conveying systems The scope of components covered in this report is as follows: • Air compressors • Air Dryers • Fly ash vacuum pumps • Buffer Hopper / Silo aeration blowers • Airlock valves • Equalizer valves • Ash storage silos • Material handling valves • Bag filters • Vent fans
4.0	REQUIREMENTS:
	Due to poor ash evacuation rate from ESP / APH / Economiser hoppers, ash accumulation in these hoppers taking place. Excess ash accumulation leads to generation loss as well as ESP internals damage. To prevent such incident, below mentioned operating procedure is to be followed. The necessary protection and interlocks of different equipments of DAES are given in Annexure-1
	The stations having dry ash hoppers evacuation and transportation by pneumatic conveying systems to be commissioned prior to ESP Pass charging and joint protocol by Operation/ C&I / Electrical Maintenance / AHP Maintenance / commissioning / OEM/ESP vendors to be made. In addition to Equipment wise commissioning, special care to be taken for commissioning of the systems as per Annexure-2
5.0	DRY ASH EVACUATION SYSTEM OPERATION PROCEDURE:
5.1	GENERAL INSTRUCTIONS:
5.1.1	Before starting the DAES, check which of the equipment is available and accordingly plan an appropriate system for transportation of dry ash.
	Resp: AHP oprn Shift Engr
5.1.2	Before starting any system, check that the equipment is not under P.T.W
	Resp: AHP oprn Shift Engr
5.1.3	Check from location that the equipment is ready for start and no persons are working on the equipment.
	Resp: AHP oprn Shift Engr
5.1.4	Good communication between the control room and different equipment locations is essential.
	Resp: SCE
5.1.5	For better equipment performance and safe operation, it is essential to ensure that the equipment is operating satisfactorily while it is in operation. This can be achieved by checking the each and every part of the system. AHP operator shall carryout detailed inspection of the equipment daily and check various points, using the format enclosed at Annexure-3 .

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	Resp: AHP oprn Shift Engr
5.1.6	To maintain DAES shift operation document, Operator has to fill Shift Log Book (e log may be used) as per Annexure-4, 4.1, 4.2 & 4.3
	Resp: AHP oprn Shift Engr
5.1.7	Based on the walk-down check-lists (Annexure-5), the operation engineer will issue work orders to the maintenance department for the repair of identified defects.
	Resp: AHP oprn Shift Engr
5.1.8	A change-over schedule is to be prepared and the equipment change-over is to be followed as per the schedule and weekly PM of Buffer Hopper to be ensured. (Annexure-6).
	Resp: AHP oprn Shift Engr /AHD/C&I
5.1.9	ASPH shift engineer shall maintain the following documentation.
	a) PTW Register indicating the equipment for which the PTW is issued and cancelled in the shift. b) Equipment availability. c) Log-book indicating the operating hours, forced outages and critical points requiring attention.
	Resp: AHP oprn Shift Engr
5.1.10	In case of any emergency defect, the AHP shift engineer shall immediately inform to concerned maintenance and also raise notification in SAP.
	Resp: AHP oprn Shift Engr
	Before Unit Startup:
	• Ensure the ESP hopper heaters are in Service. • All MHV Manual valves in full open condition. • All ESP Branches Shut-off & Line vacuum to be checked. • Buffer Hopper shut-off to be checked for air ingress. • BH Fluidizing blower started & Heaters kept in Auto. • All Valves Auto / Sequence operation to be checked. • Buffer Hopper & Silo Pulse jet Operation & pressure variation to be checked.
	Resp: AHP oprn Shift Engr
	After Unit shut down:
	• Ensure complete emptiness of all ESP hoppers. Keep the ash evacuation system in service intermittently till stopping of Fans & Rapping system. • Ensure emptiness of all Buffer hoppers & Air lock vessel. Otherwise the cold ash will cause hopper choking which becomes difficult to clear • Ensure emptiness of Silo to avoid discharge line choking • Ensure the healthiness of the system by keeping it in service at least once in a week during prolonged shutdown. • During long shut down, keep the IAC / TAC in service at least once in two days to avoid screw element jamming problem in case of screw compressor.
	Resp: AHP oprn Shift Engr
5.2	CONTROL PHILOSOPHY & SEQUENCE OF OPERATION OF DAES:
5.2.1	INSTRUMENT AIR COMPRESSOR: Reciprocating / Screw type Instrument air compressors provided for catering the requirement of instrument air of each stage units. Instrument air compressor is to be started with its required permissive before starting the ash evacuation through

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	pneumatic conveying system.
	Resp: AHP oprn Shift Engr
5.2.2	The integral controls, permissive and interlocks, load & unload operation of the instrument air compressor shall be in monitored as per OEM recommendation.
	Responsibility: SCE/C&I/EMD/AHM
5.2.3	Stage-1 is provided with 02 IACs (1W+ 1S/B) along with 02 Heater blower re-activated type air Driers. Stage-2 is provided with 04 IACs (2W+ 2S/B) along with 04 Heater blower re-activated type air Driers. Stage-1 augmentation system is provided with 02 IACs (1W+ 1S/B) along with 02 Heater blower re-activated type air Driers. The control and operation of instrument air compressor driers are from local control panel. It is to be taken in service after starting of Instrument Air Compressor.
	Resp: AHP oprn Shift Engr
5.3	DAES Operation for Vacuum Extraction:
	Control philosophy & Sequence of operation for each Vacuum Evacuation stream: The operation of fly ash is initiated from DDCMIS/OWS/LVS of AHP Control Room in the following sequence.
5.3.1	Select the Buffer hoppers for each stream of vacuum conveying from HMI if the ash level in that buffer hopper is not high.
	Resp: AHP oprn Shift Engr
5.3.2	The opening of buffer hopper isolation valves and vacuum pump seal water valve are auto based on above selection to be ensured.
	Resp: AHP oprn Shift Engr
5.3.3	Bag filter purging of selected buffer hopper will take auto start from dry mode selection. Trend of Pulse jet Air pressure to be monitored meticulously (More pressure drop indicates PJT line leakage & Less Pressure drop indicates Non- operation of solenoid valve / line chokage / Single SOV operation).
	Resp: AHP oprn Shift Engr
5.3.4	Vacuum pump of the selected buffer hopper to be started.
	Resp: AHP oprn Shift Engr
5.3.5	The buffer hopper fluidizing blower to be started. When the blower starts, the fluidizing blower heater comes into service in auto.
	Resp: AHP oprn Shift Engr
5.3.6	Once, the fluidizing blower and heater are in service, the fluidizing valves of selected Buffer hoppers are opened in auto.
	Resp: AHP oprn Shift Engr
5.3.7	The operator has to select the mode of operation (Auto/Manual) for each ESP streams from HMI.
	Resp: AHP oprn Shift Engr
5.3.8	Under auto mode, select any ESP hopper(s) for vacuum evacuation and the cleaning of various hoppers is taken up in a sequence (selected hoppers from 1st field to last field). Any MHV under maintenance to be deselected from sequence.
	Resp: AHP oprn Shift Engr
5.3.9	Under manual mode any hopper can be selected through the selection switches provided in HMI. However, only one hopper can be selected manually for ash removal

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	at a time.
	Resp: AHP oprn Shift Engr
5.3.10	Vacuum conveying system to be started through the Vacuum System Start / Stop button provided in HMI for each ESP stream. The corresponding Vacuum Breaker will close and vacuum will start developing. The opening of branch header valve of the first branch in auto with system start to be ensured.
	Resp: AHP oprn Shift Engr
5.3.11	For the auto sequence, Sequence Start / Stop buttons are provided on HMI. Sequence Start button to be pressed for initiation of auto sequence when required permissive are available.
	Resp: AHP oprn Shift Engr
5.3.12	- It is to be ensured that Material Handling Valve shall open in auto if vacuum value is greater than VS1. - During each conveying operation whenever ash is discharged into the vacuum conveying stream, the vacuum increases and when it reached the VS3 value, the MHV will close. - Shortly after the ash is conveyed, the vacuum will automatically come down and when it reaches value less than VS4, the MHV will again open and ash feeding into the system will again start. Utilizing this automatic feed control system, the ash is removed from individual hopper and the same is valid irrespective of auto or manual mode. - If the fly ash line is excessively loaded and vacuum level exceeds VS5, Vacuum Breaker will automatically open. The MHV of running hopper will again open when the vacuum value is less than VS4.
	Resp: AHP oprn Shift Engr/ AGM (C&I)
5.3.13	It is to be monitored that when the hopper is empty, ash feed into the system will be very less in quantity resulting the line vacuum coming down to low level i.e. vacuum value less than VS2. After a time lag, sequence will shift to next selected hopper and MHV of that hopper will open and the same sequence will be repeated as described above.
	Resp: AHP oprn Shift Engr
5.3.14	VS1, VS2, VS3, VS4 & VS5 are to be set by operator for Line Vacuum and total operation of vacuum conveying.
	Resp: AHP oprn Shift Engr
5.3.15	Many abnormal conditions can occur in the vacuum conveying system during deashing causing hopper/Line choking. Under such conditions, the vacuum value will remain same for a long duration. This condition is annunciated as 'Hopper Plugged'. Operator has to get it clear MHV mouth choking and ash flow to be established & hopper emptiness to be ensured. In this case, during auto cleaning mode of FA Vacuum system operation, the sequence will shift to the next selected hopper in auto.
	Resp: AHP oprn Shift Engr
5.3.16	Once the particular branch is cleaned, the operation will shift to the next branch in auto. It is to be noted that the entire auto operation will take place in sequential manner already set in the logic.
	Resp: AHP oprn Shift Engr
5.3.17	It is to be ensured that the next cycle will start automatically after completion of one sequence unless interrupted / stop from the HMI.

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	Resp: AHP oprn Shift Engr					
5.3.18	If the Dust sensor of Vacuum pump is activated, because of excessive ash ingress in vacuum line, an annunciation shall be generated in control room for necessary action for attending the bag filters.					
	Resp: AHP oprn Shift Engr/AGM (C&I)					
5.3.19	The vacuum extraction system shall be operated as per the following vacuum settings.					
	S/NO	No load vacuum mmHg (VS1)	Hopper empty vacuum mmHg (VS2)	Hopper valve intermittent close mmHg (VS3)	Hopper valve re-open mmHg (VS4)	Vacuum breaker open mmHg (VS5)
	STAGE1	100	100	220	200	450
	STAGE2	110	100	380	280	500
	Resp: AHP oprn Shift Engr/AGM (C&I)					
5.3.20	If any high level alarm is activated for ESP hopper, that particular hopper is to be evacuated in manual mode on top priority.					
	Resp: AHP oprn Shift Engr					
5.3.21	If Sequence Stop is pressed, the MHV of current hopper will close and sequence will be held at that step. It is to be ensured that sequence will again start from the same hopper when Sequence start is pressed again.					
	Resp: AHP oprn Shift Engr					
5.3.22	On system stop command, the vacuum breaker shall open and evacuation processes shall stop.					
	Resp: AHP oprn Shift Engr					
5.4	DAES Operation for Pressure Transportation: The operation of pressure conveying of fly ash to be initiated from DDCMIS / OWS / LVS of AHP Control Room in the following sequence.					
	Resp: AHP oprn Shift Engr					
5.4.1	First the selection of pressure conveying line to be done by operator by selection window available on HMI. In stage-1: Conveying line P1/P2/P3 to corresponding Silos. In stage-2: In unit-3 Conveying line 3X/3Y/3Z to any Silo and in unit-4 Conveying line 4X/4Y/4Z to any Silo as per the line & silo availability.					
	Resp: AHP oprn Shift Engr					
5.4.2	TRANSPORT AIR COMPRESSOR: Transport air compressors are provided for catering the conveying air requirement for transportation of dry fly ash from buffer hopper to silo for each unit. The TAC is to be started with its required permissive before the dry ash evacuation through pressure conveying system.					
	Resp: AHP oprn Shift Engr					
5.4.3	The overall operation of TAC. (start/stop) is initiated from HMI of DDCMIS. However, the integral controls, permissive and interlocks, load & unload operation of the Transport Air Compressor shall be as given by the OEM recommendation.					
	Resp: AHP oprn Shift Engr /C&I					
5.4.4	Refrigerant type air drier provided for transport air compressors per unit. The control and operation of transport air compressor drier is carried out through integral controller details of which are covered in the OEM control write-up.					
	Resp: AHP oprn Shift Engr /C&I					

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5.4.5	In stage-1 On each pressure conveying line, two ALVs can be selected as first vessel and second vessel. This selection to be done by operator considering the selection of Buffer hoppers (already selected in vacuum evacuation system) and pressure conveying lines as selected above. In case of St-2 each buffer hopper is having 4 ALVs and configured as pair of 1&2 or 3&4 and one pair of ALVs from a selected buffer hopper can be selected for operation. In one pressure conveying line maximum 2 pairs of ALVs from different buffer hoppers can be selected as per the availability.
	Resp: AHP oprn Shift Engr
5.4.6	Transport air compressors are provided to cater the requirement of fluidizing and transport air for pressure conveying lines of each unit.
5.4.7	The TAC discharge valves and line isolation valves are to be opened by operator through HMI to connect the working TAC to each of the pressure conveying line.
	Resp: AHP oprn Shift Engr
5.4.8	The fluidizing line valve and conveying line valves of the selected pressure conveying lines are then opened by operator through HMI.
	Resp: AHP oprn Shift Engr
5.4.9	In stage-1 dry ash can be transported to existing st-1 silos or augmentation silo through corresponding conveying lines. P2A line can be connected to any St-1 augmentationsilo from unit-2 ALVs. P3A line can be connected to any St-1 augmentation silo from unit-1 ALVs. Accordingly TACs and conveying lines are to be lined up before starting the any TAC.
	Resp: AHP oprn Shift Engr
5.4.10	Then Silo ash inlet valves to be opened for each of the pressure conveying line through HMI. It is possible to select two conveying lines of different units to the selected silo of corresponding stage.
	Resp: AHP oprn Shift Engr
5.4.11	Then, through HMI the vent fan and vent filter purging system of the silo in use to be taken in service. Purging will takes start automatically when vent fan is started. It is to be ensured.
	Resp: AHP oprn Shift Engr
5.4.12	Then the TACs to be taken in service after checking the requisite permissive. It is recommended that the selected conveying lines are to be purged with transport air for approximately 90sec before the ALVs operation is started. All the ALV valves (i.e. fluidizing air inlet valve, discharge valve, vent valve and ash inlet valve) shall be closed during line purging. Conveying line pressure to be ensured around 0.30ksc.
	Resp: AHP oprn Shift Engr
5.4.13	Once the path from transport air compressors to FA Silo is through and the required permissive are available, ALVs operation to be carried out.
	Resp: AHP oprn Shift Engr
5.4.14	It is recommended to carry out flushing of ALVs before running in normal mode where the ash loading and unloading takes place in cyclic manner.
	Resp: AHP oprn Shift Engr
5.4.15	The operation of ALVs to be started in timer based and there will be two timers (Timer-1 for first vessel and Timer-2 for second vessel) for each pressure conveying line.
5.4.16	Each ALV timer cycle comprises of filling delay time, filling time, conveying delay time, conveying time which are all user defined and will be given by operator. When the

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	system start button is pressed, both the timers initiate.
	Resp: AHP oprn Shift Engr
5.4.17	First Timer: After filling delay time, filling cycle of the first vessel starts & it's vent valve opens and after 10 seconds, the ash intake valve opens and will remain open for filling time duration. After this conveying delay timer will start. At the end of this timer, conveying timer is initiated & the fluidizing air valve opens. After 10 seconds, the ALV discharge valve opens and ash is conveyed to silo.
5.4.18	Second Timer: After conveying delay time, conveying cycle of the second vessel starts & it's fluidizing valve opens and after 10 seconds, the ash discharge valve opens and will remain open for conveying time duration. After this filling delay timer will start. At the end of this timer, filling timer is initiated & the vent valve opens. After 10 seconds, the ALV ash intake valve opens and ash is loaded into ALV. These auto operations are to be monitored.
5.4.19	The above cycle of both timers will run continuously till manually interrupted and action to be taken accordingly.
	Resp: AHP oprn Shift Engr/C&I
5.4.20	This cycle of loading of one ALV and conveying of other ALV for a pressure conveying line continues for the total period of operation of pressure conveying system. In case, the line pressure value shoots up higher than the value defined for ash conveying, the ALV discharge valves and fluidizing valves will be closed till the line pressure comes down below the set value and the cycle will start from there itself (timers will be paused). If line pressure shoots up further & TAC unloads, then TAC to be loaded if not loaded on auto when Line pressure comes down below the set value.
5.4.21	It is to be ensured that the conveying of the both vessels should not take place simultaneously.
	Resp: AHP oprn Shift Engr/C&I
5.4.22	For stopping of system, FA pressure system stop button is to be pressed and the valves of the ALV will be closed after the completion of the ongoing cycle.
5.4.23	For planned shutdown / changeover for PM, it is recommended to empty Buffer Hopper completely and that after completion of the ALV cycle the vessels are purged with air.
	Resp: AHP oprn Shift Engr
5.4.24	Silo purging system, vent fan and all the valves of silos are to be closed from HMI.
	Resp: AHP oprn Shift Engr
6	O&M GUIDE FOR DAES EQUIPMENT
6.1	AIR BLOWER FOR BUFFER HOPPER / FA SILO
6.1.1	BLOWER STARTUP CHECKLIST:
6.1.1.1	This start-up procedure should be followed during the initial installation and after any shutdown periods or after the blower has been worked on or moved to a new location. It is suggested that the steps be followed in sequence and checked off in the boxes provided.
	Resp: AGM (AHM)
6.1.1.2	Check the unit and all piping for foreign materials and clean if required.
	Resp: AGM (AHM)
6.1.1.3	Check the flatness of the feet and alignment of the drive. Feet that are bolted down

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	in a bind can cause distortion of the casing, disturbing the internal clearances.														
	Resp: AGM (AHM)														
6.1.1.4	If blowers are V belts driven, check the belt tension and alignment. Over- tensioned belts create heavy bearing loads, which leads to premature failure. Misaligned V belts can cause the impellers to rub against the side plates, resulting in overheating and jamming on operation. Misaligned couplings can cause premature bearings / shaft failures.														
	Resp: AGM (AHM)														
6.1.1.5	Be sure adequate drive guards are in place to protect the operator from severe personal injury from incidental contact.														
	Resp: AGM (AHM)														
6.1.1.6	Check the unit for proper lubrication. Proper oil level cannot be overemphasized. Too little oil will ruin bearings and gears. Too much oil will cause overheating and can ruin gears and cause other damage. Ensure both drive end & driven end are properly lubricated.														
	Resp: AGM (AHM)														
6.1.1.7	With motor locked out, turn the drive shaft by hand to be certain that impellers do not bind.														
	Resp: AGM (AHM)														
6.1.1.8	“Jog” the unit with the motor a few times to check direction of rotation and be certain it turns freely and smoothly.														
	Resp: AGM (AHM)														
6.1.1.9	Start the unit and operate for 15 minutes at no load. During this time check for abnormal noise and other indications of interference. Apply the load and observe the unit for one hour. Check frequently for abnormal noise / over heating / overloading during the first day of operation. If malfunctions occur, do not continue to operate. Problems such as knocking impellers can cause serious damage if the unit is operated without correction.														
	Resp: AGM (AHM)														
FAULT CHART															
	<table><tr><th>SYMPTOM</th><th>PROBABLE CAUSE</th><th>REMEDY</th></tr><tr><td>No air flow</td><td>Speed too low. Wrong direction of rotation. Obstruction in piping.</td><td>Check speed and verify as per recommendations. Check for correct direction of rotation. Check suction and discharge line for any obstruction</td></tr><tr><td>Low capacity</td><td>Speed too low. Excessive pressure. Discharge line leaks.</td><td>Check speed, belt drive slippage. Check the line pressure. Check up for open flow path.</td></tr><tr><td>Excessive power consumption of blower</td><td>Speed too high. Differential pressure across the suction and discharge</td><td>Correct the speed. Check line pressure with rated pressure. Check & Clean filter</td></tr></table>			SYMPTOM	PROBABLE CAUSE	REMEDY	No air flow	Speed too low. Wrong direction of rotation. Obstruction in piping.	Check speed and verify as per recommendations. Check for correct direction of rotation. Check suction and discharge line for any obstruction	Low capacity	Speed too low. Excessive pressure. Discharge line leaks.	Check speed, belt drive slippage. Check the line pressure. Check up for open flow path.	Excessive power consumption of blower	Speed too high. Differential pressure across the suction and discharge	Correct the speed. Check line pressure with rated pressure. Check & Clean filter
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		openings too high. Impellers rubbing.	and silencer. Check for any obstruction in the suction and disch. line. Check the impellers for any hot spots / rubbing marks. If observed, reset impeller timing
	Overheating of bearings and gears / blower	Inadequate lubrication. Differential pressure across the suction and discharge opening too high. Coupling misalignment. Belt slippage / over tightened belts. Too much oil in gear case. Worn impeller, clearances too high. Internal contact	Check for oil level and replace dirty oil. Check line pressure with rated pressure. Check & clean filter and silencer. Check for any obstruction in the suction and discharge line. Check alignment and realign. Adjust belt tension. Correct oil level. Replace impellers. Reset clearances.
	Vibration	Drive misalignment. Impeller rubbing (Distant knocking sound would be observed). Worn bearings / gears. Loose pulley / coupling.	Check alignment. Check for hot points. Recheck blower alignment and mounting, Recheck impeller timing. Check and replace the bearings / gears. Check if pulley / coupling is loose on shaft. Check key.
	Blower jams after running for a short period.	Insufficient axial clearances. Differential pressure across suction & discharge high, resulting in overheating & subsequent jamming, due to thermal expansion	Correct clearances. Correct differential pressures.
	Blower makes heavy knocking sound on running	Unit out of time. Distortion due to improper mounting or pipe strains. Differential pressure across	Retime impellers. Check mounting alignment and relieve pipe strains. Check gauge readings across

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		the suction and discharge openings too high. Worn bearings / gears.	the suction and discharge ends. Check and replace the bearings / gears.
	Oil leaking out of oil fill hole	Excessive oil level. Oil seal leakage.	Correct oil level. Check and replace oil seals.
	Traces of oil in blower casing.	Oil seal leakage	Check and replace oil seals.
6.2	SCREW COMPRESSOR:		
6.2.1	PRECAUTIONS DURING OPERATION:		
6.2.1.1	Never touch any piping or components of the compressor during operation.		
	Resp: AHP oprn Shift Engr		
6.2.1.2	Use only the correct type and size of hose end fittings and connections. When blowing through a hose or air line, ensure that the open end is held securely. A free end will whip and may cause injury. Make sure that a hose is fully depressurized before disconnecting it.		
	Resp: AHP oprn Shift Engr /AGM (AHM)		
6.2.1.3	Persons switching on remotely controlled machines shall take adequate precautions to ensure that there is no one checking or working on the machine. To this end, a suitable notice shall be affixed to the remote start equipment.		
	Resp: AHP oprn Shift Engr		
6.2.1.4	Never operate the machine when there is a possibility of taking in flammable or toxic fumes, vapors or particles.		
	Resp: AHP oprn Shift Engr		
6.2.1.5	Never operate the machine below or in excess of its limit ratings.		
	Resp: AHP oprn Shift Engr		
6.2.1.6	Keep all bodywork doors shut during operation. The doors may be opened for short periods only, e.g. to carry out routine checks. Wear ear protectors when opening a door. On compressors without bodywork, wear ear protection in the vicinity of the machine.		
	Resp: AHP oprn Shift Engr / AGM (AHM)		
6.2.1.7	People staying in environments or rooms where the sound pressure level reaches or exceeds 80 dB(A) shall wear ear protectors.		
	Resp: AHP oprn Shift Engr / AGM (AHM /C&I/EMD)		
6.2.1.8	Periodically check that: • All guards are in place and securely fastened. • All hoses and/or pipes inside the machine are in good condition, secure and not rubbing. • There are no leaks. • All fasteners are tight. • All electrical leads are secure and in good order. • Safety valves and other pressure relief devices are not obstructed by dirt or paint. • Air outlet valve and air net, i.e. pipes, couplings, manifolds, valves, hoses, etc. are in good repair, free of wear or abuse.		

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	Resp: AHP oprn Shift Engr /AGM (AHM/EMD)	
6.2.1.9	Do not remove or tamper with the sound-damping material.	
	Resp: AHP oprn Shift Engr /AGM (AHM)	
6.2.1.10	Never remove or tamper with the safety devices, guards or insulations fitted on the machine. Every pressure vessel or auxiliary installed outside the machine to contain air above atmospheric pressure shall be protected by a pressure relieving device or devices as required.	
	Resp: AHP oprn Shift Engr /AGM (AHM)	
6.2.2	PRECAUTIONS DURING MAINTENANCE OR REPAIR:	
6.2.2.1	Always use the correct safety equipment (such as safety glasses, gloves, safety shoes, etc.).	
	Resp: AGM (AHM/EMD/C&I)	
6.2.2.2	Use only the correct tools for maintenance and repair work.	
	Resp: AGM (AHM/EMD/C&I)	
6.2.2.3	All maintenance work shall only be undertaken when the machine has cooled down.	
	Resp: AGM (AHM/EMD/C&I)	
6.2.2.4	Close the compressor air outlet valve before connecting or disconnecting a pipe.	
	Resp: AGM (AHM/EMD/C&I)	
6.2.2.5	Before removing any pressurized component, effectively isolate the machine from all sources of pressure and relieve the entire system of pressure.	
	Resp: AGM (AHM/EMD/C&I)	
6.2.2.6	Never weld or perform any operation involving heat near the oil system. Oil tanks must be completely purged, e.g. by steam-cleaning, before carrying out such operations. Never weld on, or in any way modify, pressure vessels.	
	Resp: AGM (AHM/EMD/C&I)	
6.2.2.7	Make sure that no tools, loose parts or rags are left in or on the machine.	
	Resp: AGM (AHM/EMD/C&I)	
6.2.2.8	All regulating and safety devices shall be maintained with due care to ensure that they function properly.	
	Resp: AGM (AHM/EMD/C&I)	
6.2.3	MAINTENANCE :	
6.2.3.1	Period Running hours Operation:	
	Daily once in 8 hrs	Check the readings.
	Daily	Before starting, the level should be in the middle of the sight-glass. Check the oil level.
	Weekly	Drain condensate from the air receiver, if installed
	Every 3 months	Clean compressor.
	Every 3 months	Check for possible leaks.
	Every 3 months / 500hrs	Check the coolers; clean them if necessary.
	When displayed	Carry out service actions according to the displayed service plans.
	Resp: AHP oprn Shift Engr /AGM (AHM)	
6.2.3.2	Service Plan: A number of service operations are grouped (called level A, level B, level C, ...). Each level stands for a number of service actions to be carried out at the time intervals.	

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6.2.3.3	Servicing procedures Air filters: 1. In dusty operating conditions where it will be necessary to replace the filter too frequently, an appropriate pre-filter of the dry type should be installed. 2. The filter must be serviced as soon as the service warning appears on the display. 3. Stop the compressor, press the emergency stop button and switch off the voltage. Never remove the elements while the compressor is running. 4. Remove the filter housing cover and clean the filter chamber. Clean the cover. 5. Remove the filter. Take care that no dirt drops into the inlet pipe. Clean and inspect the element sealing surface of the filter chamber. 6. Install the new or cleaned filter and fit the cover. Never install damaged or clogged filters. 7. Switch on the voltage, unlock the emergency stop button and reset the air filter service warning.			
	Resp: AGM (AHM)			
6.2.3.4	OIL AND OIL FILTER CHANGE :			
	Procedure:			
6.2.3.4.1	Run the compressor until warm.			
	Resp: AHP oprn Shift Engr /AGM (AHM)			
6.2.3.4.2	Stop the compressor, press the emergency stop button.			
	Resp: AHP oprn Shift Engr /AGM (AHM)			
6.2.3.4.3	Switch off the voltage.			
	Resp: AHP oprn Shift Engr /AGM (AHM)			
6.2.3.4.4	The filler plug and drain points are painted yellow. Remove filler plug (FC). Drain the compressor sump by opening drain valve (Dm1). Drain the oil cooler by Opening its drain valve. Close the drain valves and tighten the drain plugs after draining.			
	Resp: AHP oprn Shift Engr /AGM (AHM)			
6.2.3.4.5	Remove the oil filters (OF). Clean the filter seats, oil the gaskets of the new filters and screw the filters into place until the gaskets contact the seats. Then tighten by hand.			
	Resp: AHP oprn Shift Engr /AGM (AHM)			
6.2.3.4.6	Fill the compressor sump to the middle of the oil level sight-glass (SG). Use oil as per OEM recommendation.			
	Resp: AHP oprn Shift Engr /AGM (AHM)			
6.2.3.4.7	Reinstall the filler plug.			
	Resp: AHP oprn Shift Engr /AGM (AHM)			
6.2.3.4.8	Switch on the voltage, unlock the emergency stop button and reset the oil and oil filter service warnings.			
	Resp: AHP oprn Shift Engr /AGM (AHM)			
6.2.3.4.9	Run the compressor for a few minutes.			
	Resp: AHP oprn Shift Engr /AGM (AHM)			
6.2.3.4.10	Stop the compressor, check that the oil level is in the middle of sight-glass (SG).			
	Resp: AHP oprn Shift Engr /AGM (AHM)			
6.2.3.4.11	Top up, if necessary.			
	Resp: AHP oprn Shift Engr /AGM (AHM)			
	FAULTS AND REMEDIES:			
	Condition	Fault	Remedy	

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	Oil pressure too low	Oil level too low	Top up level to the middle of the oil level sight-glass
	Oil temperature above normal	Oil shut-down temperature incorrectly set	Correct setting to trip at the specified temperature
		Insufficient cooling water flow or water inlet temperature too high	Check and increase cooling water flow or decrease water inlet temperature, if possible
	Compressor capacity or working pressure lower than normal	Air consumption exceeds capacity of compressor	Check pneumatic plant
		Safety valves leaking	Remove leaking valve and have it inspected
		On compressors equipped with a full-load/no-load regulator also: Unloading pressure incorrectly set	Correct as necessary
		Resp: AGM (AHM)	
6.3	REFRIGERATION DRYER:		
6.3.1	MAINTENANCE: Maintenance schedule of refrigeration type Dryer		
	Daily	Drain the condensate from the moisture separator and air receiver.	
			Resp: AHP oprn Shift Engr
	Weekly	Working of compressor	
			Resp: AGM (TMD)
	Monthly	Mounting bolts of the compressor, condenser. De scaling of water cooled condenser.	
			Resp: AGM (TMD)
	Yearly	Quantity of refrigerant gas	
			Resp: AGM (TMD)
6.3.2	MAINTENANCE OF REFRIGERATION DRYER: (air cooled)		
	Daily	Open the manual drain valve for one second every 3 hours of working so as to check if any condensates discharge itself.	
			Resp: AHP oprn Shift Engr
		Normally no condensate should discharge since it is automatically being discharged through the auto drain. Excessive discharge from manual drain indicates clogged orifice of the auto drain which requires cleaning. The orifice can be cleaned only when the air is shut off.	
			Resp: AHP oprn Shift Engr
		The following is the procedure for cleaning the orifice.	
		Discharge through manual drain.	
		Close ball valve of auto-drain valve.	
		Disconnect nylon tubing from the bottom of the separator.	
		Disconnect the solenoid valve and clean with air pressure.	
		Connect the valve and tube. And open ball valve of auto-drain valve.	

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		Resp: AGM (TMD)
	Weekly	Since this machine has an air cooled condenser, care should be taken that mud should not be allowed to clog the fins of condenser coil. This will lower the efficiency of the machine and shall cause the dew point to rise. The condenser should be periodically cleaned.
		Clean the fan blade periodically
		Check the refrigerant through the sight glass. The appearance of the bubbles during normal running of the Dryer indicates shortage of the gas. (call for authorized service engineer of the company.)
	Yearly	Check the mounting bolts of the compressor, receiver, condenser and liquid Separator for tightness.
		Resp: AGM (TMD)
6.4	VENT FAN	
6.4.1	PRECAUTIONS BEFORE STARTING: • Check the mounting of the bearing housings and electric motor. • Check that the coupling is tightened properly. • Check that the foundation bolts are properly tightened. • Check that the connecting flanges between the fan and ducting are good, and proper packing is provided. • Check that the different components of the control system are correctly connected. • Check that the Inlet damper is fully closed. • Check that no foreign object has been left anywhere inside the fan. • Check that no water stays in the bottom of the casing and inlet box. • Check that the specified guards have been fitted. • Check that the inspection door on the casing and inlet box are correctly shut. • Check that the rotor can turn freely without rubbing with any static parts. • Check that the bearings have been correctly filled with bearing lubricant as it is noted in the section "LUBRICATION".	
		Resp: AHP oprn Shift Engr
6.4.2	VENT FAN MAINTENANCE:	
6.4.2	RUNNING MAINTENANCE: 1. Fill bearing lubricant of recommended specification and quantity at the specified intervals for bearing housings. The interval should be determined based on local condition like working environment etc. 2. Check periodically vibration levels of the fan at both bearings. 3. Check periodically the proper seating of fan on the foundation and for abnormal vibration at any location. 4. Check for looseness of any bolts in the fan pedestals, casing, etc.	
		Resp: AHP oprn Shift Engr : AGM (AHM)/MTP
6.4.2.2	SHUT DOWN MAINTENANCE: • Open inspection door in casing. Clean the impeller blades and internals. • Check for erosion of the impeller. • Check the overlap and radial clearance between impeller shroud and inlet cone. If found disturbed adjust the same. • Remove used bearing lubricant from bearing housing.	

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	• Check the bearing condition like roller clearance and the rolling surfaces of races. • Fill new lubricant in the bearing housing. • Check alignment of fan and motor. If necessary correct the same. • Check the Inlet damper flaps movement. • Apply grease for Inlet damper bearing housings. • Check the tightness of all fixing bolts
	Resp: AGM (AHM)
6.5	VENT FILTER
6.5.1	START-UP, TIMING ADJUSTMENTS AND STOPPAGE: When the bag filter is erected, it must first be checked to ensure that all components are correctly mounted and respect to the conditions normally prevailing during the operation but these checks and adjustments can be performed in accordance with the guide.
	Check and Ensure: • That no tools or other objects are left inside the bag filter or inside Fan casing. • That filter bags are not damaged and mounted in correct position. • That the inspection door/s is properly closed. • That the solenoid valves are installed correctly. • That all the bearings are greased properly i.e. with correct grade and quantity. • That the fan inlet damper is in closed condition. • That all electrical connections are made properly. • That the centrifugal fan rotates in the right direction.
	Resp: AHP oprn Shift Engr : AGM (AHM/C&I/ EMD)
	START-UP: Sequence of operations: • Check air pressure in the compressed air tank and if it is found lower than the specified rate, then start the compressor and allow time to build up specified air pressure more than 5.4kg/cm² in the compressed air tank. • Allow 10 sec. to clear up the system. • Start the Electronic Controller of bag filter. • Allow 5 min. For bag cleaning. • Start Fan motor and wait till it reaches full speed. • Slowly open the fan inlet damper to full open position.
	Resp: AHP oprn Shift Engr /C&I
	STOPPAGE : Sequence of operation: • Stop the Centrifugal fan and close the inlet damper. • Allow 5 min. Time for bag cleaning. • Switch off power supply of electronic controller and compressor.
	Resp: AHP oprn Shift Engr
6.6	VACUUM PUMP
6.6.1	VACCUM PUMP (04 WEEKLY PM) • Check/top up oil level in bearing housing. • Check/top up oil level in bearing gear box. • Check tightness of belts & foundation bolts. • Cleaning of suction filters. • Gland checking

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	Resp: AGM (AHM)
6.6.2	VACUUM PUMP OVERHAUL
6.6.2.1	BEARING Lubrication: • Bearings are to be lubricated in every six months. • Remove bearing caps for checking • Inspect the condition and quality of grease inside the bearing housing • If quality of grease is not satisfactory, pack the bearings with quality grease until the grease is flush with the bearing's inner & outer race. • Pack each bearing cap.
	Resp: AGM (AHM)
6.6.2.2	STUFFING BOX PACKING: • Remove gland cover • Remove old gland packing • Insert new & fresh gland packing. The cut ring has to be slant. The cut surface of each ring has to be placed opposite to each other. • Carefully tighten the gland cover.
	Resp: AGM (AHM)
6.6.2.3	DISMENTALING OF VACUUM PUMP: If a disassembly of the pump is required for replacement of pump spares, proceed the followings:- • Remove suction & discharge piping. • Remove the Belts, Pulley & coupling from the shaft. • Remove suction & discharge piping. • Remove the Belts, Pulley & coupling from the shaft. • Remove the bearing brackets and bearings from both sides. • Remove the old gland packing from the stuffing box. • Remove the nuts of the casing cover. • Remove the impeller with shaft from casing to remove casing cover with control cone on one side. • Remove another casing cover with control cone. • Using the allen key, remove the control cone from casing cover.
	Resp: AGM (AHM)
6.6.2.4	INSPECTION OF VACUUM PUMP PARTS: With pump disassembled, inspect parts for wear as described as following:- CASING: a) Clearance to be checked between shaft & casing. b) Check for uneven wear on the casing inner surface. c) If minimum clearance is more than 1mm, replace casing CONTROL CONES: a) Check wear and scoring between the ports and at the edges of the ports with a straightedge. b) If wear is more than 3mm, replace the cone. IMPELLER: a) Inspect the control cone bore of the impeller on the basis as the surface of the control cone. b) Check for uneven wear, undercutting. c) If wear is more than 1mm, replace the impeller.

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	SHAFT: - check for uneven wear and scoring on the shaft surface. - Replace shaft if it is not repairable.
	Resp: AGM (AHM)
6.6.2.5	ASSEMBLY OF VACUUM PUMP: - Match the control cone in the casing cover and tighten. - Fit the casing cover with the casing at one side. - Insert the impeller with shaft in the casing. - Fit the casing cover with control cone at another side. - Locate impeller position in center by axial movement. - Fit the gland packing in the stuffing box gland at both end. - Fit the bearing bracket and bearing at both ends. - Fit the pulley/coupling on the shaft. - Make alignment on the belt/coupling and connect the pump to the system.
	Resp: AGM (AHM)
6.7.1	BUFFER HOPPERS (01 WEEKLY PM) AIR LOCK VALVE ASSEMBLY: - Check smoothness & completeness of operation. - Check condition of valve seat, disc & body for wear / replace the worn out components - Check actuator shaft/adaptor for play, erosion/wear & replace the worn out components. - Attend leakages from loose or missing fasteners.
	Resp: AGM (AHM)
	EQUALIZING VALVE ASSEMBLY: - Check smoothness & completeness of operation - Arrest leak from adaptor cover, gland packing. - Check valve spindle/gland follower for wear & replace if required. Apply small qty. Of molykote/ polykote to spindle. - Adjust v/v operation speed control for stroke - Clean all moving parts as required. - Arrest gasket leaks if hiss sound or material out. - Check equalizing valve lines for leakages & attend.
	Resp: AGM (AHM)
	BUFFER HOPPERS (04 WEEKLY (BREAKDOWN/OVERHAULING)) AIR LOCK VALVE ASSEMBLY - Perform 01 weekly checks. - Lubricate shaft adaptor with molykote / polykote. - Check aeration stone/pads for damage / choking / air leakage & replace / repair the damaged components.
	Resp: AGM (AHM)
	EQUALIZING VALVE ASSEMBLY - Perform 01 weekly checks. - Equalizing valve disc & seat must from air tight seal.
	Resp: AGM (AHM)
	BUFFER HOPPER / AIR LOCK TANKS: Check all round & square aeration pads for damage/choking/air leakages &

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	replace/repair the damaged components. • Arrest leakages from buffer hopper manholes & ash inlet pipe adopter couplings. • Inspect bag filters/cages if ash discharge is reported from vac. Pumpsoutlet & replace bag filter set if required.
	Resp: AGM (AHM)
	MATERIAL HANDLING VALVES (04 WEEKLY PM): • Check for loose/missing fasteners • Check for any air leakage & tighten gland packing, as required. • Check seat and slide, replace if found eroded.
	Resp: AGM (AHM)
	SEGREGATING VALVES (04 WEEKLY PM): • Check smoothness & completeness of operation. • Attend leakages from gland, end cap, access cover etc. • Check valve sealing and pneumatic air leakage. • Check seat and slide, replace if found eroded.
	Resp: AGM (AHM)
7.0	RESPONSIBILITIES The Head of the Operation & AHP Maintenance Group, will provide guidance regarding the interpretation of the technical features of this document.
8.0	REVIEW: The document will be reviewed 2 yearly basis or as and when required.

ANNEXURE-1

Permissive: Fly ash system
1. Vacuum pumps:-
1. Water line valve open & wetting unit Pr. Available $>3.5 \text{ kg/cm}^2$ 2. Suction valve open 3. Any FAHP running 4. Vacuum breaker not closed & line valve open 5. Ash inlet valve open 6. EPB released 7. Feeder healthy 8. Over load not there 9. Seal water pressure low $<1.5 \text{ kg/cm}^2$.
2. Vacuum pump seal water valve: -
a) Open permissive with open feedback of corresponding vacuum pump suction valve. b) Indications in HMI: open/ close status.
3. Vacuum breaker:-
a) Close permissive: - selected vacuum pump is running and corresponding vacuum path is through. b) Protection Open: - when close permissive are not available. Actuations:-i) Open and Close buttons in HMI. ii) Auto close pulse with Vacuum system start button. iii) Auto open pulse if vacuum rises above VS5 value.
4. Material handling valve:-
a) Open permissive:- i) Vacuum pump of selected buffer hopper is running. ii) Selected vacuum breaker closed. iii) Pulse jetting of selected buffer hopper is ON. iv) Selected bag filter Diff pr not high. v) Any fluidizing blower is ON. vi) Vacuum with operating range (i.e. vacuum more than VS1 & less than VS3). vii) Corresponding branch header valve is open. viii) Hopper is not bypassed. ix) Selected buffer hopper level is not high. If any of the above permissive are not available the MHV will close in auto. b) Actuations:- i) Open and close buttons in HMI. ii) Auto open / close command from auto / manual sequence. C) Indications in HMI: open/ close status.
5. Branch header valve:-
a. Open permissive: - Any buffer hopper selected. b. Actuations:- i) Open & close buttons in HMI. ii) Auto open pulse in manual mode – if corresponding branch is selected. iii) Auto open pulse in auto mode with vacuum system start button- Only applicable for first branch header valve of each pass. iv) Auto open / close command from vacuum conveying system auto sequence.

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c. Indications in HMI: open/ close status.
6. Buffer hopper fluidizing blower:- a) Start permissive: - Nil b) Actuations: - Start and stop buttons in HMI. c) Indications in HMI: ON/OFF status.
7. Buffer hopper fluidizing Heater:- a. Start permissive: - corresponding blower ON and temp less than set point. Heater banks will trip if corresponding blower is off or temp is more than set point. b. Actuations:- I. Start and stop buttons in HMI. II. Auto cut in and cut off – based on temp set point. c. Indications in HMI: ON/OFF status, heater temp analog value.
8. Vacuum system auto sequence (separate sequence for each ESP pass):- a) Sequence start permissive:- i) Instrument air pressure is normal. ii) Vacuum pump of selected buffer hopper is running. iii) Selected vacuum breaker closed. Pulse jetting of selected buffer hopper is ON. iv) Any fluidizing blower is ON. v) No load vacuum is established (i.e. vacuum more than VS1). vi) Selected buffer hopper level is not high. b) Actuations:- i) Auto sequence Start and stop buttons in HMI. ii) When Auto sequence stop button is pressed, the sequence will hold at that step and selected MHV will close and when auto sequence start is pressed again sequence will start from step where it was held. iii) Indications in HMI: current cycle completed, number of cycles counter, Time taken by current cycle. Vacuum set points, vacuum analog value.
9. Pressure system auto sequence (separate sequence for each pressure conveying line):- a) Sequence start permissive:- i) Pressure Conveying line through. ii) TAC running. iii) Normal/Flushing Mode Selected. iv) START Push Button pressed. b) Actuations: - Auto sequence Start and stop buttons in HMI.

Annexure-2**Commissioning of DAES:**

Sl No	Description				
	The major instrumentations, its settings, interlocks & protections are to be commissioned as follows:-				
1.	Instrument air pressure should be $> 5.4 \text{ kg/cm}^2$ for cleaning of Buffer Hopper & Silos Bag filters (Pulse Jet Diaphragm opening on timer 50-80msec , 12sec off timer interval)				
2.	Buffer Hoppers ALV inlet & outlet valves limit switches must be commissioned. In case of Dome valves, seal pr. must be 2 kg/cm^2 more than conveying air pr. & seal pr. switch must not be by-passed.				
3.	Sequence operation & Interlocks on above valves must be in service.				
	• Vent valve opens 10sec. before opening of ALV intake valve. • ALV intake valve opens for 75% of ALV filling (either level mode filling or timer setting, to be set) • Closing of ALV ash intake & vent valve. • If ALV inlet valve not closed, ALV discharge valve will not open. • When ALV vent valve closed, ALV pressure equalizing valve opens. • ALV fluidizing solenoid opens 10sec before opening of ALV discharge valve. • ALV discharge valve opens for preset time period when other ALV discharge valve in same FA line is closed.				
4.	Commissioning of Buffer Hopper ash level hi. During trial operation, same to be ensured.				
5.	Buffer Hopper bag filter DP in service (for protection of bag filters)				
6.	MHV closing on the above to be ensured(to avoid ash accumulation in Buffer Hopper) & start with same MHV after reset of permissive				
7.	Pr. Gauges to be installed before & after TA line orifice/regulating valve to ensure designed DP against orifice/regulator: 0.25 kg/cm^2				
8.	Pr. Transmitter to be commissioned before orifice (in lean ph. Sys), after orifice (in dense ph. Sys). Its reading must be available at control room. All Fly Ash Line settings to be commissioned for the same.				
	<table border="1"> <tr> <td>FA line pr. Hi</td><td>Dumping stopped & MHV closed.</td></tr> <tr> <td>FA line pr. low</td><td>Dumping starts & MHV opens.</td></tr> </table>	FA line pr. Hi	Dumping stopped & MHV closed.	FA line pr. low	Dumping starts & MHV opens.
FA line pr. Hi	Dumping stopped & MHV closed.				
FA line pr. low	Dumping starts & MHV opens.				
9.	Buffer Hopper vacuum breaker must open immediate with the stopping of Vacuum Pump. This is for protection of bag filters from vacuum Pump seal water ingress.				
10.	The vacuum extraction system shall be operated as per the following vacuum settings. These settings are tentative and shall be set at site as per actual arrangement. • No load vacuum 140 mmHg (VS1). • Hopper empty 100 mmHg (VS2). • Hopper valve intermittent close on Hi vacuum, $300\text{-}350 \text{ mmHg}$ (VS3) to avoid line blockage • Hopper valve re-open $250\text{-}300 \text{ mmHg}$ (VS4) to maintaining conveying phase density • Vacuum breaker open on Hi-Hi Vacuum, 450 mmHg (VS5) • Vacuum breaker close 250 mmHg. (VS6)				
11.	N.B. Setting of (VS3) & (VS4) is very vital as conveying phase density depends on it. Higher				

Operation & Maintenance Procedure for Dry Ash Evacuation & Transportation System

	(VS3) setting increases phase density but decreases velocity which should not be less than pick-up velocity. Lower setting of (VS4) increases line purging but decreases conveying phase density.
12.	Individual Buffer Hopper fluidizing airline orifice as per design to be provided to distribute equal flow/pr. In all passes, A to E. Solenoids also to be provided to control flow at the time of gate operation only to maintain pr.
13.	Fluidizing air heater to be commissioned with auto cut-out at 120°C & cut-in at 80°C.
14.	Vacuum Pump current to be displayed at control room (Current <110/120amps at 350mmHg vacuum, seal water flow 50/65 lit/min, 12/15mm orifice at 2.5kg/cm ² for Vacuum Pump 3000/3900m ³ /hr).
15.	Display of Vacuum transport line temperature may be explored to ensure ash flow.
16.	TAC driers to be commissioned & DP of dryers within design value
17.	Commissioning/establishment of APH/Duct hoppers evacuation
18.	Ensuring ECO flush water pressure at flushing apparatus and establishment of ECO hoppers evacuation.

Annexure-3

DAES OPERATION CHECK LIST

A VACUUM PUMP IN SERVICE							
		Unit	A1/1A2	B1/1B2	C1/1C2	D1/1D2	E1/1E2
1	SEALING WATER PRESSURE	OK / Not OK					
2	VENT WATER QUALITY HAVING ASH	Yes/No					
3	BELT CONDITION	OK / Not OK					
4	LINE VACUUM	MMHg					
6	SHUTOFF VACUUM REACHING TIME IN SEC	sec					
7	VACUUM PUMP CURRENT AT 350MMHG VAC.	Amps					
B BUFFER HOPPER IN SERVICE							
		Unit	A	B	C	D	E
1	INSTRUMENT AIR PRESSURE	Kg/cm ²					
2	PULSE JET AIR PRESSURE	Kg/cm ²					
3	BUFFER HOPPER BAG FILTER DP	MMWC					
4	OBSERVE & RECORD BRANCH NOT PULSATING	Pulsating / Not pulsating					
5	REPORT AIR INGRESS SOUND IF ANY	Yes/No					
6	SHUT OFF VACUUM	MMHg					
7	SEQUENCE OF ALV VALVES OPERATION	OK/Not OK					
C SILO AERATION BLOWER							
		Unit	1	2	3	4	
1	CHECK OIL LEVEL	OK/Not OK					
2	ABNORMAL SOUND FROM BLOWER	Yes/No					
3	OUTLET AIR TEMP. AFTER HEATER	°C					
4	BELT CONDITION	OK/Not OK					
D TRANSPORT AIR COMPRESSOR IN SERVICE							
		Unit	TAC-1	TAC-2	TAC-3		
1	OUTLET AIR TEMP. AFTER COOLER	°C					
2	LUB OIL TEMPERATURE	°C					
3	LUB OIL PRESSURE	°C					
4	ABNORMAL SOUND FROM COMP. IF ANY	YES/NO					
5	DRAIN THE MOISTURE FROM ALL POINTS	YES/NO					
6	BACK FLUSH COOLER IF REQUIRED	YES/NO					
7	CHECK AIR FILTER DP	mmwc					
8	COOLING WATER	Kg/cm ²					

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	PRESSURE						
9	AIR OUTLET TEMP. FROM COMPRESSOR	°C					
10	LUB OIL LEVEL	OK/ NOT Ok					
E	TAC DRIER IN SERVICE						
		Unit	TAC-1	TAC-2	TAC-3		
1	TEMP. OF REFRIGERANT COND. INLET	°C					
2	TEMP. OF REFRIGERANT COND. OUTLET	°C					
3	DEW POINT TEMP. OF AIR	°C					
4	AUTO DRAIN TRAP WORKING	YES/NO					
5	MANUALLY DRAIN THE MOISTURE	Yes / No					
F	DRIER REGENERATION BLOWER IN SERVICE						
		Unit	IAC-1	IAC-2	IAC-3	IAC-4	
1	AIR TEMP. AT OUTLET	°C					
2	ANY ABNORMAL SOUND	YES/NO					
3	RAISE DEFECT IF DISCHARGE AIR TEMP. >45°C	YES/NO					
H	SILO IN SERVICE						
		Unit	1	2	3		
1	SILO VENT FAN IN SERVICE	YES/NO					
2	PULSE JET INSTRUMENT AIR PR.	Kg/cm ²					
3	PULSE JET SEQUANCE	OK/ NOT					
4	ANY LEAKAGE ON SILO TOP	YES/NO					

Operation & Maintenance Procedure for Dry Ash Evacuation & Transportation System

ANNEXURE-4

ASH HANDLING DESK OPERATOR'S LOG BOOK

UNIT		GROUP		SHIFT		DATE		LOAD		COAL FLOW			
SHIFT BEGINING STATUS													
VACCUM PUMPS I/S / AVL				A1	A2	B1	B2	C1	C2	D1	D2	E1	E2
WETTING UNITS I/S / AVL				A1	A2	B1	B2	C1	C2	D1	D2	E1	E2
ESP FIELD STATUS :													
STREAM	ABOVE LOW LEVEL					HIGH LEVEL		HOPPERS CLEARED					
A													
B													
C													
D													
E1													
E2													
STREAMS OPEN BREAKER / SHUT OFF / HEADER VACUUM LEVELS & CURRENT													
STREAMS	A1 / A2		B1 / B2		C1 / C2		D1 / D2		E1		E2		
	VACUUM MMHg	CURT. A	VACUUM MMHg	CURT. A	VACUUM MMHg	CURT. A	VACUUM MMHg	CURT. A	VACUUM MMHg	CURT. A	VACUUM MMHg	CURT. A	
OPEN BR.													
SHUT OFF													
HDR VAC	1												
	2												
	3												
	4												
DRY ASH EVACUATION :													
SHIFT BEGINING STATUS : ALV IN DRY MODE						SHIFT ENDING STATUS : ALV IN DRY MODE							
P1	1A1	1B1	2A1	2B1			P1	1A1	1B1	2A1	2B1		
P2	1A2	1B2	1C1	1D1	1E1		P2	1A2	1B2	1C1	1D1	1E1	
	2A2	2B2	2C1	2D1	2E1			2A2	2B2	2C1	2D1	2E1	
P3	1C2	1D2	1E2	2C2	2D2	2E2	P3	1C2	1D2	1E2	2C2	2D2	2E2
SHIFT ENDING STATUS:													
VACCUM PUMPS I/S / AVL				A1	A2	B1	B2	C1	C2	D1	D2	E1	E2
WETTING UNITS I/S / AVL				A1	A2	B1	B2	C1	C2	D1	D2	E1	E2
STREAM	ABOVE LOW LEVEL					HIGH LEVEL		EQUIPMENTS UNDER PTW / NA :					
A													
B													
C													
D													
E1													
E2								DESK OPERATORS NAME :					
								SIGNATURE :					

Annexure-5

DRY FLY ASH EVACUATION SYSTEM WALK DOWN CHECKLIST

Sl No	DESCRIPTION	Remarks / Checked Status		
		SHIFT-A	SHIFT-B	SHIFT-C
	ASPH			
1	Check for oil leakage from gear box / hydraulic coupling.			
2	Check for any abnormal sound / vibration of the pump or Motor.			
3	Check for any pump gland / gasket / body leakage.			
4	Check for any belt slipping / smoke.			
5	Check for any Ash slurry sump overflow.			
6	Check for lighting inside / outside pump house.			
7	Check for any unsafe condition in the ASPH.			
	AWPH			
1	Check for any abnormal sound / vibration of the pump or Motor.			
2	Check for any pump gland / gasket / body leakage.			
3	Check running pumps bearing temperatures.			
4	Check for closing of all standby pump discharge valves.			
5	Check for any standby pump NRV passing / reverse rotation if any.			
6	Check for any Ash water tank overflow.			
7	Check for lighting inside / outside pump house.			
8	Check for any unsafe condition in the AWPB.			
	ESP HOPPERS / BUFFER HOPPERS / ALVs			
1	Check for any spillage of ash while poking of hoppers.			
2	Check for any ash leakage from ESP hoppers / pressure conveying lines / ALVs / Buffer hoppers.			
3	Check for any instrument air leakage			
4	Check for ESP header air intake valve healthiness in st-1 and orifice size according vacuum pump in service (Big or small) in st-2.			
5	Check for any air ingress in the vacuum system couplings.			
6	Check for any passing of MHV / header valves			
7	Check for wetting unit water pressure.			
8	Check for bag filter purging system solenoid valves sequential operation and Instrument air pressure			
9	Check for closing of standby Wetting Units water valve.			
10	Check for lighting in ESP /BH/ALVs area.			
11	Check for any unsafe condition in ESP /BH /ALVs area.			
12	Check Instrument air pressure at pulse jet station			
13	Check Fluidizing Line charged and Temperature is OK.			

Operation & Maintenance Procedure for Dry Ash Evacuation & Transportation System

14	ALV Fluidizing Line valve Auto operation.			
	ESP VACUUM PUMPS			
1	Check for any abnormal sound / vibration of the pump or Motor.			
2	Check for any vacuum pump belt slipping.			
3	Check for any water leakage from the vacuum pump.			
4	Check for Vacuum pump seal water pressure (Between 1 to 1.5 ksc)			
5	Check for closing of standby vacuum pump seal water.			
6	Check for any ash coming from vacuum pump discharge water.			
7	Check for Local vacuum in pr gauge / transmitter.			
8	Check for lighting in vacuum pumps area.			
9	Check for any unsafe condition in ESP Vacuum pumps area.			
10	Check for Air ingress sound if any			
11	Instrument / Transport air leakages nearby if any			
	TACs & DRYER			
1	Check for TAC / dryer gear box oil level			
2	Check for any abnormal sound from compressor / motor			
3	Check for any oil / air leakage from TAC.			
4	Check for TAC air receiver auto solenoid operation / any drain line chocking			
5	Check for TAC motor winding temp, bearing temperature and oil temperature and compressor discharge pressure in Local panel .			
6	Check for dryer local parameters like suction gas pressure / discharge pressure, Dew point temperature etc at local panel.			
7	Check for healthiness of dryer moisture drain.			
8	Check for lighting in TAC and dryer area.			
9	Check for any unsafe condition in TAC & Dryer area.			
10	Check for Cooling water Pressure/flow			
11	Check for Instrument air leakages if any.			
	Aeration Blower			
1	Check Suction filter differential pressure			
2	Check Outlet temp. of air/Heater working or not			
3	Check Oil level .			
4	Check for Abnormal vibration if any			
5	Check Bearing temperature.			
6	Check for Leakage of air if any			
	ASH Silo			
1	Check for Ash leakage if any.			
2	Check for Water ingress point if any			
3	Check for Air leakage if any.			
4	Check Sequence of operation of pulse jet system at running Silo.			
5	Pr. Drop of pulse jet within limit			

Operation & Maintenance Procedure for Dry Ash Evacuation & Transportation System

6	Check Silo vent fan abnormal sound if any.			
7	Check SVF Belt condition/ Bearing temperature within limit.			
8	Check Dust from Vent fan discharge			
9	Check for Instrument air pressure at pulse jet station within limit			
	IAC & DRYER			
1	Check for IAC gear box oil level			
2	Check for any abnormal sound from compressor / motor			
3	Check for any oil / air leakage from IAC.			
4	Check for IAC air receiver auto solenoid operation / any drain line chocking			
5	Check for dryer local parameters like Tower in service, Tower in regeneration, Blower in service, Heater in service etc.			
6	Check for lighting in IAC and dryer area.			
7	Check for any unsafe condition in IAC & Dryer area.			
8	Check for Belt condition/ tension.			
9	Check for Compressor vibration.			
	Conveying lines			
1	Approach/Condition of line for welded clamps.			
2	Ash leakage from KGV & flanges.			
3	Check for Erosion of lines/ bends. Clamp looseness. Coupling de-shaped			
	CERM /EERM/HOPPER HEATERS (Local)			
1	Check for any continuous operation of CERM / EERM (Normal ON time is 1minute)			
2	Check for CERM /EERM gear box oil level			
3	Check for any abnormal sound from GB / motor / from internal.			
4	Check for hopper heaters local panel healthiness and for ON /OFF indication			
5	Check for lighting in IAC and dryer area.			
6	Check for any unsafe condition in CERM / EERM / Hopper heaters area.			
	CERM /EERM/HOPPER HEATERS (Switch Gear)			
1	Check for auto position of selection switch in CERM / EERM /GDRM in switch gear module.			
2	Check for any continuous operation of CERM / EERM in switch gear module (Normal ON time is 1minute)			
3	Check for hopper heaters ON indication and current in switch gear module			
4	Check for lighting in switch gear room.			

ANNEXURE-6**ASH HANDLING PLANT EQUIPMENT CHANGEOVER SCHEDULE**

MONDAY		Date:		
Shift	Equipment	From	To	Remarks
A	IAC & Drier St-1 : St-2: St-1 augmentation:			
	Pass-A ESP vacuum pumps Unit-1: Unit-2: Unit-3: Unit-4:			
	TAC & Dryer Unit-3: St-1 (Existing old):			
B	LP seal water pump St-1: St-2:			
	HP seal water pump St-1: St-2:			
	Silo Dust conditioner Pump St-1 : St-2: St-1 augmentation:			
TUESDAY		Date:		
Shift	Equipment	From	To	Remarks
A	Fly ash HP pump St-1: St-2:			
	Pass-B ESP vacuum pumps Unit-1: Unit-2: Unit-3: Unit-4:			
	TAC & Dryer Unit-4: St-1 / U-1 Augmentation: St-1 / U-2 Augmentation:			
B	Silo Fluidizing blower St-1: St-2: St-1 augmentation:			
WEDNESDAY		Date:		
Shift	Equipment	From	To	Remarks

Operation & Maintenance Procedure for Dry Ash Evacuation & Transportation System

A	Bottom ash HP pump St-1: St-2:			
	Pass-C ESP vacuum pumps Unit-1: Unit-2: Unit-3: Unit-4:			
B	Silo area drain sump pump St-1: St-2:			
	Unit-3 BH fluidizing blower & heater			
	Unit-3 ESP fluidizing blower & heater			
THURSDAY		Date:		
Shift	Equipment	From	To	Remarks
A	Bottom ash LP pump St-1: St-2:			
	Pass-D ESP vacuum pumps Unit-1: Unit-2: Unit-3: Unit-4:			
	Silo area wash water pumps St-1 St-2 St-1 augmentation			
	Unit-4 BH fluidizing blower & heater			
B	Unit-4 ESP fluidizing blower & heater			
FRIDAY		Date:		
Shift	Equipment	From	To	Remarks
A	Flushing Water Pump (St-1)			
	Pass-E ESP vacuum pumps Unit-1: Unit-2:			
	St-1 Vent Fan St-1 Vent Fan St-1 Vent Fan	Silo-1 Silo-2 Silo-3		
	Ash Slurry Drain sump pump St-1: St-2:			
SATURDAY		Date:		
Shift	Equipment	From	To	Remarks
A	Unit-1/2 vacuum pump drain pumps			
	St-1 aug CW system CT fans			
	St-1 aug CW system cooling water pumps			

ANNEXURE-4.1

ASH HANDLING PLANT (St-2)			
UNIT:	SHIFT:	GROUP:	DATE:
STAGE-II ESP HOPPER EVACUATION STATUS IN AUTO MODE (Put tick mark)			

Branch No	HPR No	Pass A			Branch No	HPR No	Pass B			Branch No	HPR No	Pass C			Branch No	HPR No	Pass D		
		Cycle-1	Cycle-2	Cycle-3			Cycle-1	Cycle-2	Cycle-3			Cycle-1	Cycle-2	Cycle-3			Cycle-1	Cycle-2	Cycle-3
1	H1				1	H1				1	H1				1	H1			
	H2					H2					H2					H2			
	H3					H3					H3					H3			
	H4					H4					H4					H4			
	H5					H5					H5					H5			
	H6					H6					H6					H6			
	H7					H7					H7					H7			
	H8					H8					H8					H8			
	H9					H9					H9					H9			
	H10					H10					H10					H10			
2	H11				2	H11				2	H11				2	H11			
	H12					H12					H12					H12			
	H13					H13					H13					H13			
	H14					H14					H14					H14			
	H15					H15					H15					H15			
	H16					H16					H16					H16			
	H17					H17					H17					H17			
	H18					H18					H18					H18			
	H19					H19					H19					H19			
	H20					H20					H20					H20			

Operation & Maintenance Procedure for Dry Ash Evacuation & Transportation System

Branch No	HPR No	Pass A			Branch No	HPR No	Pass B			Branch No	HPR No	Pass C			Branch No	HPR No	Pass D		
		Cycle-1	Cycle-2	Cycle-3			Cycle-1	Cycle-2	Cycle-3			Cycle-1	Cycle-2	Cycle-3			Cycle-1	Cycle-2	Cycle-3
3	H21				3	H21				3	H21				3	H21			
	H22					H22					H22					H22			
	H23					H23					H23					H23			
	H24					H24					H24					H24			
	H25					H25					H25					H25			
	H26					H26					H26					H26			
	H27					H27					H27					H27			
	H28					H28					H28					H28			
	H29					H29					H29					H29			
	H30					H30					H30					H30			
4	H31				4	H31				4	H31				4	H31			
	H32					H32					H32					H32			
	H33					H33					H33					H33			
	H34					H34					H34					H34			
	H35					H35					H35					H35			
	H36					H36					H36					H36			
	H37					H37					H37					H37			
	H38					H38					H38					H38			
	H39					H39					H39					H39			
	H40					H40					H40					H40			
	NO OF CYCLE					NO OF CYCLE					NO OF CYCLE					NO OF CYCLE			

Shift end report :		Group		Date	
UNIT No					
DEASHING DONE(Y/N)					
NO OF CYCLE					
Equipment under Permit					
Name :		Signature:			

ASH HANDLING PLANT (St-1)																			
UNIT:					SHIFT:					GROUP:					DATE:				
STAGE-1 ESP HOPPER STATUS (Put tick mark)																			
Branch No	HPR No	Pass A			Branch No	HPR No	Pass B			Branch No	HPR No	Pass C			Branch No	HPR No	Pass D		
		Cycle-1	Cycle-2	Cycle-3			Cycle-1	Cycle-2	Cycle-3			Cycle-1	Cycle-2	Cycle-3			Cycle-1	Cycle-2	Cycle-3
1	H101				5	H105				9	H109				13	H113			
	H201					H205					H209					H213			
	H301					H305					H309					H313			
	H401					H405					H409					H413			
	H501					H505					H509					H513			
	H601					H605					H609					H613			
	H701					H705					H709					H713			
	H801					H805					H809					H813			
2	H102				6	H106				10	H110				14	H114			
	H202					H206					H210					H214			
	H302					H306					H310					H314			
	H402					H406					H410					H414			
	H502					H506					H510					H514			
	H602					H606					H610					H614			
	H702					H706					H710					H714			
	H802					H806					H810					H814			
3	H103				7	H107				11	H111				15	H115			
	H203					H207					H211					H215			
	H303					H307					H311					H315			
	H403					H407					H411					H415			
	H503					H507					H511					H515			
	H603					H607					H611					H615			
	H703					H707					H711					H715			
	H803					H807					H811					H815			

Operation & Maintenance Procedure for Dry Ash Evacuation & Transportation System

Branch No	HPR No	Pass A			Branch No	HPR No	Pass B			Branch No	HPR No	Pass C			Branch No	HPR No	Pass D		
		Cycle-1	Cycle-2	Cycle-3			Cycle-1	Cycle-2	Cycle-3			Cycle-1	Cycle-2	Cycle-3			Cycle-1	Cycle-2	Cycle-3
4	H104				8	H108				12	H112				16	H116			
	H204					H208					H212					H216			
	H304					H308					H312					H316			
	H404					H408					H412					H416			
	H504					H508					H512					H516			
	H604					H608					H612					H616			
	H704					H708					H712					H716			
	H804					H808					H812					H816			

Shift end report :		Group		Date	
UNIT No					
DEASHING DONE(Y/N)					
NO OF CYCLE					
Equipment under Permit					
Name :		Signature:			

ANNEXURE -4.2**ESP FLY ASH EVACUATION HOPPER SELECTION STATUS IN MANUAL MODE****UNIT - I**

Dt:				Shift:A/B/C				Group-1/2/3/4											
Pass-A				Pass-B				Pass-C				Pass-D							
A1		A2		B1		B2		C1		C2		D1		D2					
VP I/S: A1/A2				VP I/S: B1/B2				VP I/S: E1/E2				VP I/S: C1/C2				VP I/S: D1/D2			
101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116				
201	202	203	204	205	206	207	208	209	210	211	212	213	214	215	216				
301	302	303	304	305	306	307	308	309	310	311	312	313	314	315	316				
401	402	403	404	405	406	407	408	409	410	411	412	413	414	415	416				
501	502	503	504	505	506	507	508	509	510	511	512	513	514	515	516				
601	602	603	604	605	606	607	608	609	610	611	612	613	614	615	616				
701	702	703	704	705	706	707	708	709	710	711	712	713	714	715	716				
801	802	803	804	805	806	807	808	809	810	811	812	813	814	815	816				

ESP FLY ASH EVACUATION HOPPER SELECTION STATUS IN MANUAL MODE**UNIT - II**

Dt:				Shift:A/B/C				Group-1/2/3/4							
Pass-A				Pass-B				Pass-C				Pass-D			
A1		A2		B1		B2		C1		C2		D1		D2	
VP I/S: A1/A2			VP I/S: B1/B2			VP I/S: E1		VP I/S: E2		VP I/S: C1/C2			VP I/S: D1/D2		
101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116
201	202	203	204	205	206	207	208	209	210	211	212	213	214	215	216
301	302	303	304	305	306	307	308	309	310	311	312	313	314	315	316
401	402	403	404	405	406	407	408	409	410	411	412	413	414	415	416
501	502	503	504	505	506	507	508	509	510	511	512	513	514	515	516
601	602	603	604	605	606	607	608	609	610	611	612	613	614	615	616
701	702	703	704	705	706	707	708	709	710	711	712	713	714	715	716
801	802	803	804	805	806	807	808	809	810	811	812	813	814	815	816

ESP FLY ASH EVACUATION HOPPER SELECTION STATUS IN MANUAL MODE**UNIT - III****Dt:****Shift:A/B/C****Group-1/2/3/4**

Pass-A				Pass-B				Pass-C				Pass-D			
A1		A2		B1		B2		C1		C2		D1		D2	
VP I/S: A1/A2				VP I/S: B1/B2				VP I/S:C1/C2				VP I/S:D1/D2			
1	11	21	31	1	11	21	31	1	11	21	31	1	11	21	31
2	12	22	32	2	12	22	32	2	12	22	32	2	12	22	32
3	13	23	33	3	13	23	33	3	13	23	33	3	13	23	33
4	14	24	34	4	14	24	34	4	14	24	34	4	14	24	34
5	15	25	35	5	15	25	35	5	15	25	35	5	15	25	35
6	16	26	36	6	16	26	36	6	16	26	36	6	16	26	36
7	17	27	37	7	17	27	37	7	17	27	37	7	17	27	37
8	18	28	38	8	18	28	38	8	18	28	38	8	18	28	38
9	19	29	39	9	19	29	39	9	19	29	39	9	19	29	39
10	20	30	40	10	20	30	40	10	20	30	40	10	20	30	40

ESP FLY ASH EVACUATION HOPPER SELECTION STATUS IN MANUAL MODE**UNIT - IV****Dt:****Shift:A/B/C****Group-1/2/3/4**

Pass-A				Pass-B				Pass-C				Pass-D			
A1		A2		B1		B2		C1		C2		D1		D2	
VP I/S:A1/A2				VP I/S:B1/B2				VP I/S:C1/C2				VP I/S:D1/D2			
1	11	21	31	1	11	21	31	1	11	21	31	1	11	21	31
2	12	22	32	2	12	22	32	2	12	22	32	2	12	22	32
3	13	23	33	3	13	23	33	3	13	23	33	3	13	23	33
4	14	24	34	4	14	24	34	4	14	24	34	4	14	24	34
5	15	25	35	5	15	25	35	5	15	25	35	5	15	25	35
6	16	26	36	6	16	26	36	6	16	26	36	6	16	26	36
7	17	27	37	7	17	27	37	7	17	27	37	7	17	27	37
8	18	28	38	8	18	28	38	8	18	28	38	8	18	28	38
9	19	29	39	9	19	29	39	9	19	29	39	9	19	29	39
10	20	30	40	10	20	30	40	10	20	30	40	10	20	30	40

ANNEXURE-4.3

NTPC - SIMHADRI, STAGE-I (2 × 500 MW UNITS)
ESP HOPPER HEATERS CHECKING LOG SHEET

UNIT- I						Group:						Date:						Shift:					
PASS- A						PASS- B																	
Hopper No	HH NO	Current (Amps) R/Y/B	Hopper No	HH NO	Current (Amps) R/Y/B	Hopper No	HH NO	Current (Amps) R/Y/B	Hopper No	HH NO	Current (Amps) R/Y/B												
101 & 102	1A11		103 & 104	1A21		105 & 106	1B11		107 & 108	1B21													
201 & 202	2A11		203 & 204	2A21		205 & 206	2B11		207 & 208	2B21													
301 & 302	3A11		303 & 304	3A21		305 & 306	3B11		307 & 308	3B21													
401 & 402	4A11		403 & 404	4A21		405 & 406	4B11		407 & 408	4B21													
501 & 502	5A11		503 & 504	5A21		505 & 506	5B11		507 & 508	5B21													
601 & 602	6A11		603 & 604	6A21		605 & 606	6B11		607 & 608	6B21													
701 & 702	7A11		703 & 704	7A21		705 & 706	7B11		707 & 708	7B21													
801 & 802	8A11		803 & 804	8A21		805 & 806	8B11		807 & 808	8B21													
PASS- C						PASS- D																	
Hopper No	HH NO	Current (Amps) R/Y/B	Hopper No	HH NO	Current (Amps) R/Y/B	Hopper No	HH NO	Current (Amps) R/Y/B	Hopper No	HH NO	Current (Amps) R/Y/B												
109 & 110	1C11		111 & 112	1C21		113 & 114	1D11		115 & 116	1D21													
209 & 210	2C11		211 & 212	2C21		213 & 214	2D11		215 & 216	2D21													
309 & 310	3C11		311 & 312	3C21		313 & 314	3D11		315 & 316	3D21													
409 & 410	4C11		411 & 412	4C21		413 & 414	4D11		415 & 416	4D21													
509 & 510	5C11		511 & 512	5C21		513 & 514	5D11		515 & 516	5D21													
609 & 610	6C11		611 & 612	6C21		613 & 614	6D11		615 & 616	6D21													
709 & 710	7C11		711 & 712	7C21		713 & 714	7D11		715 & 716	7D21													
809 & 810	8C11		811 & 812	8C21		813 & 814	8D11		815 & 816	8D21													

NTPC - SIMHADRI, STAGE-I (2 × 500 MW UNITS)

ESP HOPPER HEATERS CHECKING LOG SHEET

UNIT-II		Group:				Date:				Shift:	
PASS- A						PASS- B					
Hopper No	HH NO	Current (Amps) R/Y/B	Hopper No	HH NO	Current (Amps) R/Y/B	Hopper No	HH NO	Current (Amps) R/Y/B	Hopper No	HH NO	Current (Amps) R/Y/B
101 & 102	1A12		103 & 104	1A22		105 & 106	1B12		107 & 108	1B22	
201 & 202	2A12		203 & 204	2A22		205 & 206	2B12		207 & 208	2B22	
301 & 302	3A12		303 & 304	3A22		305 & 306	3B12		307 & 308	3B22	
401 & 402	4A12		403 & 404	4A22		405 & 406	4B12		407 & 408	4B22	
501 & 502	5A12		503 & 504	5A22		505 & 506	5B12		507 & 508	5B22	
601 & 602	6A12		603 & 604	6A22		605 & 606	6B12		607 & 608	6B22	
701 & 702	7A12		703 & 704	7A22		705 & 706	7B12		707 & 708	7B22	
801 & 802	8A12		803 & 804	8A22		805 & 806	8B12		807 & 808	8B22	
PASS- C						PASS- D					
Hopper No	HH NO	Current (Amps) R/Y/B	Hopper No	HH NO	Current (Amps) R/Y/B	Hopper No	HH NO	Current (Amps) R/Y/B	Hopper No	HH NO	Current (Amps) R/Y/B
109 & 110	1C12		111 & 112	1C22		113 & 114	1D12		115 & 116	1D22	
209 & 210	2C12		211 & 212	2C22		213 & 214	2D12		215 & 216	2D22	
309 & 310	3C12		311 & 312	3C22		313 & 314	3D12		315 & 316	3D22	
409 & 410	4C12		411 & 412	4C22		413 & 414	4D12		415 & 416	4D22	
509 & 510	5C12		511 & 512	5C22		513 & 514	5D12		515 & 516	5D22	
609 & 610	6C12		611 & 612	6C22		613 & 614	6D12		615 & 616	6D22	
709 & 710	7C12		711 & 712	7C22		713 & 714	7D12		715 & 716	7D22	
809 & 810	8C12		811 & 812	8C22		813 & 814	8D12		815 & 816	8D22	

NTPC - SIMHADRI, STAGE-II (2 × 500 MW UNITS)
ESP HOPPER HEATERS CHECKING LOG SHEET

UNIT-III						Date:		Shift:			
PASS- A						PASS- B					
Hopper No	HH NO	Current (Amps) R/Y/B	Hopper No	HH NO	Current (Amps) R/Y/B	Hopper No	HH NO	Current (Amps) R/Y/B	Hopper No	HH NO	Current (Amps) R/Y/B
1&11	10		21&31	20		1&11	10		21&31	20	
2&12	11		22&32	21		2&12	11		22&32	21	
3&13	12		23&33	22		3&13	12		23&33	22	
4&14	13		24&34	23		4&14	13		24&34	23	
5&15	14		25&35	24		5&15	14		25&35	24	
6&16	15		26&36	25		6&16	15		26&36	25	
7&17	16		27&37	26		7&17	16		27&37	26	
8&18	17		28&38	27		8&18	17		28&38	27	
9&19	18		29&39	28		9&19	18		29&39	28	
10&20	19		30&40	29		10&20	19		30&40	29	
PASS- C						PASS- D					
Hopper No	HH NO	Current (Amps) R/Y/B	Hopper No	HH NO	Current (Amps) R/Y/B	Hopper No	HH NO	Current (Amps) R/Y/B	Hopper No	HH NO	Current (Amps) R/Y/B
1&11	10		21&31	20		1&11	10		21&31	20	
2&12	11		22&32	21		2&12	11		22&32	21	
3&13	12		23&33	22		3&13	12		23&33	22	
4&14	13		24&34	23		4&14	13		24&34	23	
5&15	14		25&35	24		5&15	14		25&35	24	
6&16	15		26&36	25		6&16	15		26&36	25	
7&17	16		27&37	26		7&17	16		27&37	26	
8&18	17		28&38	27		8&18	17		28&38	27	
9&19	18		29&39	28		9&19	18		29&39	28	
10&20	19		30&40	29		10&20	19		30&40	29	

NTPC - SIMHADRI, STAGE-II (2 × 500 MW UNITS)
ESP HOPPER HEATERS CHECKING LOG SHEET

UNIT-IV						Date:		Shift:			
PASS- A						PASS- B					
Hopper No	HH NO	Current (Amps) R/Y/B	Hopper No	HH NO	Current (Amps) R/Y/B	Hopper No	HH NO	Current (Amps) R/Y/B	Hopper No	HH NO	Current (Amps) R/Y/B
1&11	1		21&31	1		1&11	1		21&31	1	
2&12	2		22&32	2		2&12	2		22&32	2	
3&13	3		23&33	3		3&13	3		23&33	3	
4&14	4		24&34	4		4&14	4		24&34	4	
5&15	5		25&35	5		5&15	5		25&35	5	
6&16	6		26&36	6		6&16	6		26&36	6	
7&17	7		27&37	7		7&17	7		27&37	7	
8&18	8		28&38	8		8&18	8		28&38	8	
9&19	9		29&39	9		9&19	9		29&39	9	
10&20	10		30&40	10		10&20	10		30&40	10	
PASS- C						PASS- D					
Hopper No	HH NO	Current (Amps) R/Y/B	Hopper No	HH NO	Current (Amps) R/Y/B	Hopper No	HH NO	Current (Amps) R/Y/B	Hopper No	HH NO	Current (Amps) R/Y/B
1&11	1		21&31	1		1&11	1		21&31	1	
2&12	2		22&32	2		2&12	2		22&32	2	
3&13	3		23&33	3		3&13	3		23&33	3	
4&14	4		24&34	4		4&14	4		24&34	4	
5&15	5		25&35	5		5&15	5		25&35	5	
6&16	6		26&36	6		6&16	6		26&36	6	
7&17	7		27&37	7		7&17	7		27&37	7	
8&18	8		28&38	8		8&18	8		28&38	8	
9&19	9		29&39	9		9&19	9		29&39	9	
10&20	10		30&40	10		10&20	10		30&40	10	